April 2027

MILESTONE 1 : DATA EXPLORATION

INTRODUCTION

Facial recognition systems are widely used in modern applications such as security and identity verification. However, recognizing faces in real-world conditions is challenging due to variations in lighting, pose, and background.

In this project, the goal is to build a facial recognition system starting from data preparation. This milestone focuses on exploring the dataset and applying preprocessing techniques to make it suitable for machine learning models.

Data quality plays a critical role in model performance, making this step essential before training.

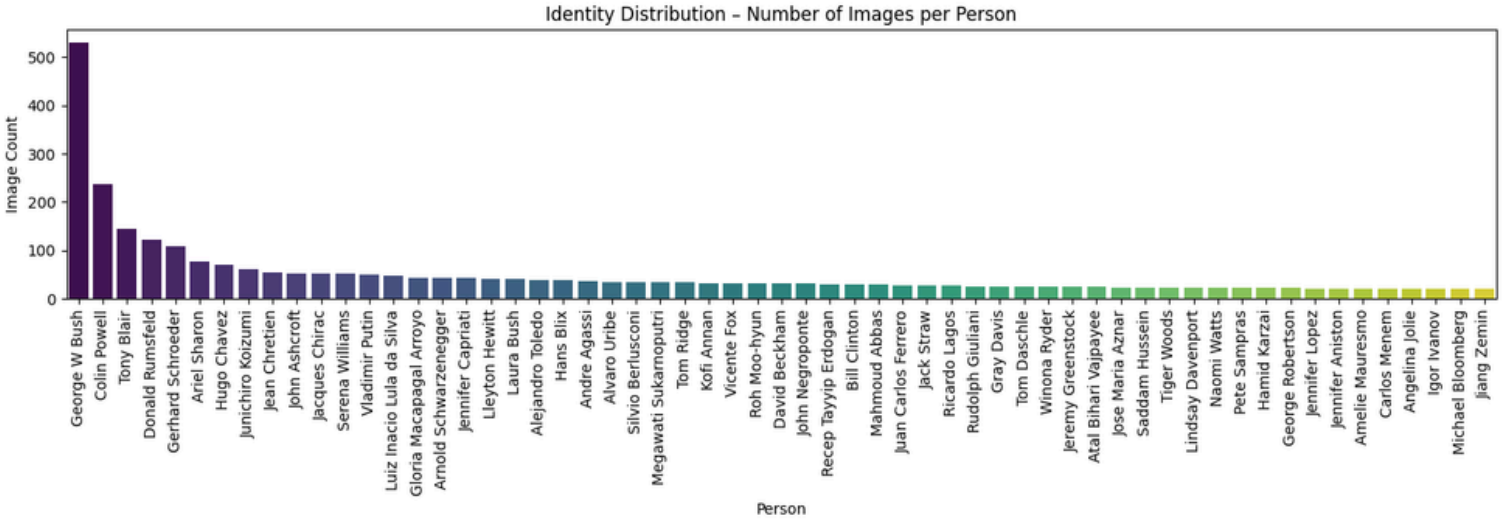
DATASET DESCRIPTION

The dataset used is Labeled Faces in the Wild **(LFW)**, which contains real-world face images collected under unideal situations:
- **Number of images: 2936**
- **Number of individuals: 59**

The dataset includes variations in lighting, pose, and facial expressions, making it suitable for real-world applications.

DATA EXPLORATION

DISTRIBUTION



The distribution of images per person is highly imbalanced.

A small number of individuals have many images, while most have few.

This may lead to biased model performance toward frequently appearing individuals.

SAMPLE IMAGES

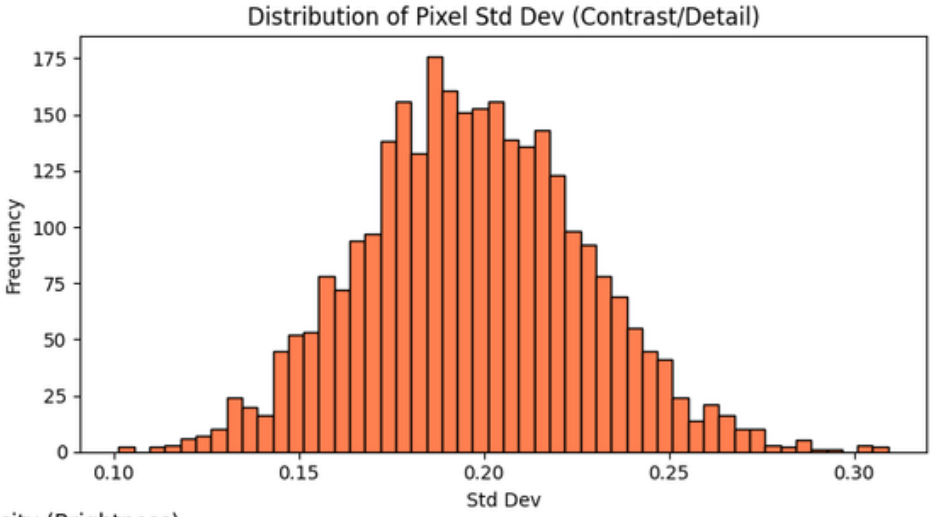
Sample images show variations in lighting, pose, and background.
Some images are clear, while others include noise or occlusion.

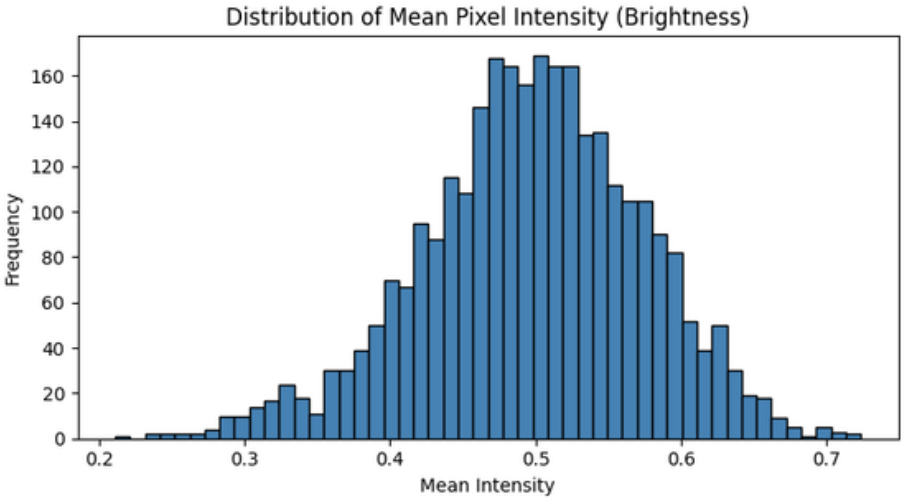
This increases the difficulty of the recognition task.

BRIGHTNESS / ANALYSIS

The brightness of images varies significantly, indicating inconsistent lighting conditions.

This can affect feature extraction and model accuracy.





CHALLENGES

Key challenges identified:

- Class imbalance
- Lighting variation
- Pose variation

These challenges require proper preprocessing to ensure robust model performance.

PREPROCESSING

- Face Detection

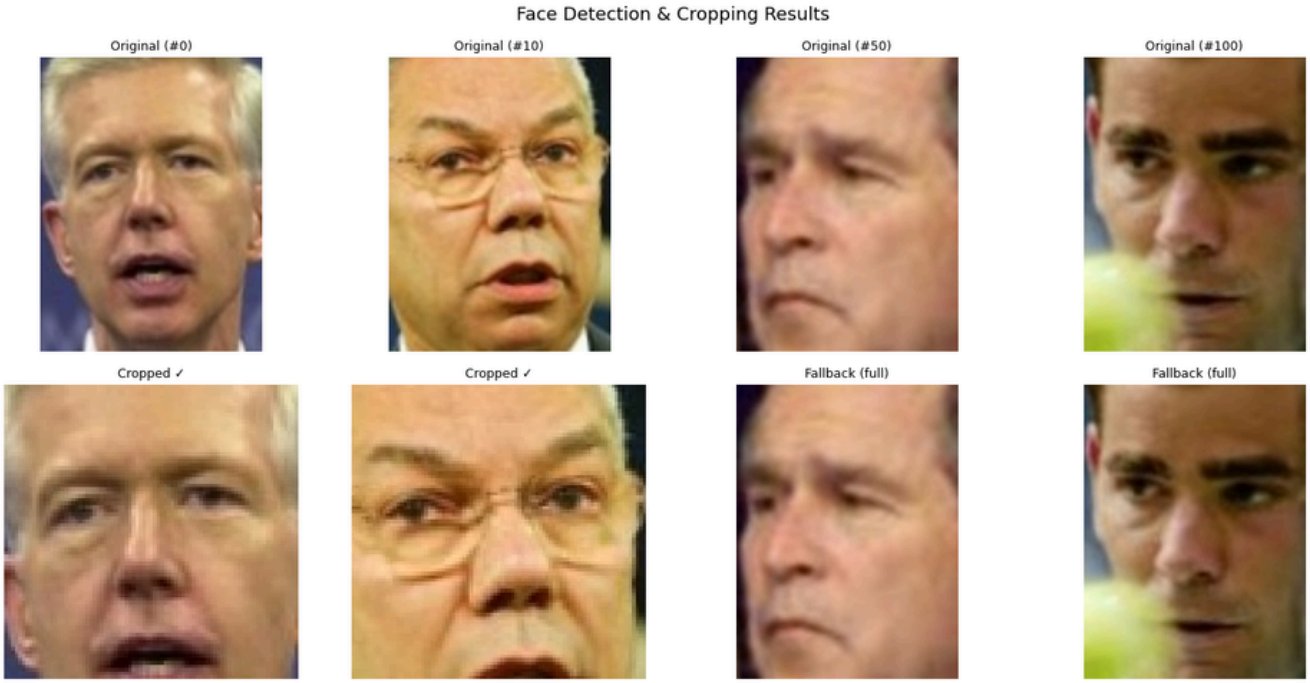
Faces were detected using Haar Cascade classifier to focus on relevant regions.

- Cropping

The detected face region was cropped from the original image.

– Resizing
Images were resized to a fixed size **(224×224)** to standardize input.

– Normalization
Pixel values were normalized to improve training performance.



DATA AUGMENTATION

Data augmentation techniques such as flipping and rotation were applied to increase dataset diversity.

This helps reduce overfitting and improves generalization.

OUTPUT DATASET

The processed dataset was saved in (.npz) format for efficient reuse during training.

This ensures faster loading and reproducibility.

CONCLUSION

In this milestone, the dataset was explored and preprocessed to prepare it for model training. Key challenges such as imbalance and lighting variation were identified and addressed.

The dataset is now ready for the next phase, which involves building and training a facial recognition model.

Summary

Item	Details
Dataset	LFW (Labeled Faces in the Wild)
Total Images	~2,936 (with ≥20 per person)
Identities	59 distinct people
Original Resolution	125×94 pixels (RGB)
Target Resolution	224×224 pixels
Normalization	Pixel values scaled to [0, 1]
Face Detection	OpenCV Haar Cascade (with fallback for undetected faces)
Augmentation	Rotation, flip, zoom, shift, brightness via Keras `ImageDataGenerator`